Soo Min Kwon

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EDUCATION

University of Michigan Sept. 2022 – June 2026
Ph.D., Electrical Engineering and Computer Science Ann Arbor, MI

- **Thesis**: "Deep Learning through Low-Dimensional Representations: Theory and Algorithms"
- **Advisor(s)**: Prof. Laura Balzano and Prof. Qing Qu

Rutgers University Sept. 2016 – May 2022
M.S., B.S., Electrical and Computer Engineering New Brunswick, NJ

- **Thesis**: "Optimization Problems with Low-Dimensional Tensor Structures"
- **Advisor**: Prof. Anand D. Sarwate

EXPERIENCE

Microsoft June 2026 – Present
Member of Technical Staff Mountain View, CA

- Integrating and improving frontier AI models into the PowerPoint Copilot agent, driving measurable gains in visual quality, instruction following, editing accuracy, and overall presentation readiness
- Used data-driven methods and statistical testing to add new features to the PowerPoint AI agent, reducing the thumbs-down rate (TDR) by 5%

Google DeepMind July 2025 – Nov. 2025
Student Researcher New York, NY

- Hosted by Himanshu Jain and Ziteng Sun in the BigML team (now the GenAI team)
- Developed CoDistill-GRPO, a novel variant of GRPO aimed at improving small language model performance via co-distillation training with larger language models
- Tested the effectiveness of CoDistill-GRPO on the Qwen and Llama model families and with models up to 8B parameters, demonstrating improvement in accuracy compared to GRPO and baselines on mathematical reasoning tasks

Amazon July 2024 – Nov. 2024
Applied Scientist Intern Seattle, WA

- Developed a causal inference framework using deep learning methods for the SCOT team that reduced variance estimates of average treatment effects by over 10%

LinkedIn May 2022 – Aug. 2022
Applied Research Scientist Intern Sunnyvale, CA

- Productionized a machine learning pipeline for the infrastructure team, reducing MAPE by over 15% in forecasting hardware needs for the next calendar year

SELECTED PUBLICATIONS († denotes equal contribution)

Generative AI (LLMs, Diffusion Models)

[1] **CoDistill-GRPO: A Co-Distillation Recipe for Efficient Group Relative Policy Optimization** [Paper]
S. M. Kwon, *H. Jain, Z. Sun, A. T. Suresh, S. Kumar*
ICML 2026 AdaptFM Workshop

[2] **The Effect of Training Task Diversity on In-Context Learning through the Lens of Low-Dimensional Subspaces** [Paper]
S. M. Kwon, *AS Xu, C Yaras, D Song, L Balzano, Q Qu*

[3] **An Overview of Low-Rank Structures in the Training and Adaptation of Large Models: From Implicit Low-Dimensionality to Efficient Training and Fine-Tuning** [Paper]
L Balzano, T Ding, BD Haeffele, ***S. M. Kwon****, Q. Qu, P Wang, Z Wang, C. Yaras (α-β Order)*
IEEE Signal Processing Magazine

[4] **Out-of-Distribution Generalization of In-Context Learning: A Low-Dimensional Subspace Perspective** [Paper]
S. M. Kwon†, *A. S. Xu*†, *C. Yaras, L. Balzano, Q. Qu*
AISTATS 2026; AISTATS 2026 CauScale Workshop **(Best Paper Award)**; DeepMath 2025 **(Oral)**

[5] **Decoupled Data Consistency with Diffusion Purification for Image Restoration** [Paper]
X Li, SM Kwon, IR Alkhouri, S Ravishankar, Q Qu
ICCP 2026 **(Oral)**

[6] **BLAST: Block-Level Adaptive Structured Matrices for Efficient DNN Inference** [Paper]
C. Lee, ***S. M. Kwon****, Q. Qu, H. Lee*
NeurIPS 2024

[7] **Solving Inverse Problems with Latent Diffusion Models via Hard Data Consistency** [Paper]
B. Song†, ***S. M. Kwon***†, *Z. Zhang, X. Hu, Q. Qu, L. Shen*
ICLR 2024 **(Spotlight, Top 5%)**

Optimization & Deep Learning Theory

[1] **Dynamic Subspace Estimation from Undersampled Data using Grassmannian Geodesics** [Paper]
S. M. Kwon*, C. Blocker, H. Raja, J. Fessler, L. Balzano*
TMLR 2026

[2] **Learning Dynamics of Deep Matrix Factorization Beyond the Edge of Stability** [Paper]
A. Ghosh†, ***S. M. Kwon***†, *R. Wang, S. Ravishankar, Q. Qu*
ICLR 2025

[3] **Efficient Compression of Overparameterized Deep Models through Low-Dimensional Learning Dynamics** [Paper]
S. M. Kwon*, Z. Zhang, D. Song, L. Balzano, Q. Qu*
AISTATS 2024

AWARDS & HONORS

Best Paper Award at the AISTATS 2026 CauScale Workshop	2026
Harvey G. and Joyce H. Behner Graduate Fellowship	2024
University of Michigan PhD Rackham Merit Fellowship	2023
Rutgers ECE Outstanding Master's Student Award	2022
Rutgers ECE Outstanding Teaching Assistant Award	2021
Rutgers ECE Departmental Leadership & Service Award	2020
Rutgers WINLAB GA/TA Grant	2020
Rutgers University Dean's List	2018 – 2020

INVITED TALKS

2025 INFORMS Annual Meeting	2025
Math Machine Learning Seminar @ MPI MIS + UCLA	2025